P2.12

Use operator representation

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a}_+ + \hat{a}_-) \qquad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}\,(\hat{a}_+ - \hat{a}_-)$$

Then, in the n-th stationary state $\psi_n(x)$,

$$\langle x\rangle = \sqrt{\frac{\hbar}{2m\omega}} \int \psi_n^* (\hat{a}_+ + \hat{a}_-)\psi_n \, dx =$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \int \psi_n^* \left(\sqrt{n+1}\,\psi_{n+1} + \sqrt{n}\,\psi_{n-1}\right) dx = 0$$ by ortogonality of ψ_n and ψ_m with $n \neq m$.

$\langle p\rangle = 0$ by the same reason.

But $$\langle x^2\rangle = \frac{\hbar}{2m\omega} \int \psi_n^* (\hat{a}_+ + \hat{a}_-)^2 \psi_n \, dx =$$

$$= \frac{\hbar}{2m\omega} \int \psi_n^* \left(a_+a_+ + a_-a_- + a_+a_- + a_-a_+\right)\psi_n =$$

$$= \frac{\hbar}{2m\omega} \int \psi_n^* \left(\sqrt{(n+1)(n+2)}\,\psi_{n+2} + \sqrt{n(n-1)}\,\psi_{n-2} + n\psi_n + (n+1)\psi_n\right) dx$$

$$= \frac{\hbar}{2m\omega}(2n+1) \int \psi_n^* \psi_n \, dx = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)$$

$$\langle p^2\rangle = -\frac{\hbar m\omega}{2} \int \psi_n^* \left(\sqrt{(n+1)(n+2)}\,\psi_{n+2} + \sqrt{n(n-1)}\,\psi_{n-2} - (2n+1)\psi_n\right) dx = \hbar m\omega\left(n+\frac{1}{2}\right)$$

$$\langle T\rangle = \frac{\langle \hat{p}^2\rangle}{2m} = \frac{\hbar m\omega}{2m}\left(n+\frac{1}{2}\right) = \frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right)$$

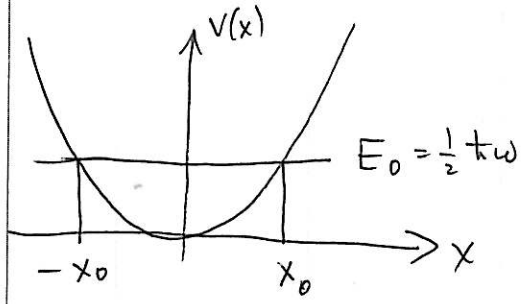
$$\langle V\rangle = \frac{1}{2}m\omega^2\langle x^2\rangle = \frac{1}{2}m\omega^2\frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right) = \frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right)$$

$$\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \sqrt{\frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)}, \quad \sigma_p = \sqrt{\hbar m\omega\left(n+\frac{1}{2}\right)}$$

$$\Rightarrow \sigma_x\sigma_p = \hbar\left(n+\frac{1}{2}\right) \geq \frac{\hbar}{2}$$ OK

P2.15

$$\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2} \qquad E_0 = \frac{1}{2}\hbar\omega$$



Classical turning point x_0:

$$\frac{1}{2}\hbar\omega = \frac{1}{2} m\omega^2 x_0^2$$

$$\Rightarrow x_0 = \pm\sqrt{\frac{\hbar}{m\omega}}$$

The probability to find particle <u>outside</u> $(-x_0, x_0)$ interval is:

$$P = 2\int_{x_0}^{\infty} |\psi_0|^2 dx = 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \int_{x_0}^{\infty} e^{-\frac{m\omega}{\hbar}x^2} dx$$

Change variables to $\xi = \sqrt{\frac{m\omega}{\hbar}}\, x \Rightarrow x = \sqrt{\frac{\hbar}{m\omega}}\,\xi$

$$\Rightarrow \xi_0 = \sqrt{\frac{m\omega}{\hbar}}\, x_0 = \pm 1.$$

$$\Rightarrow P = 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\frac{\hbar}{m\omega}\right)^{1/2} \int_1^{\infty} e^{-\xi^2} d\xi$$

In terms of <u>error</u> function $\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2} dt$

$$P = \frac{2}{\sqrt{\pi}}\left(\int_0^{\infty} e^{-\xi^2} d\xi - \int_0^1 e^{-\xi^2} d\xi\right) = \mathrm{erf}(\infty) - \mathrm{erf}(1)$$

But $\mathrm{erf}(\infty) = \frac{2}{\sqrt{\pi}}\int_0^{\infty} e^{-\xi^2} d\xi = \frac{2}{\sqrt{\pi}}\,\frac{1}{2}\int_{-\infty}^{\infty} e^{-\xi^2} d\xi = 1$

$$\Rightarrow P = 1 - \mathrm{erf}(1) = 1 - 0.843 = 0.157$$

$$P = 0.157.$$

P2.16 Eq. (2.84) $$a_{j+2} = \frac{-2(n-j)}{(j+1)(j+2)} a_j$$

$\underline{n=5}$ (odd $\Rightarrow h_{even} = 0 \Rightarrow a_0 = a_2 = a_{even} = 0$).

$j=1 \quad a_3 = \frac{-2(5-1)}{2\cdot 3} a_1 = -\frac{4}{3} a_1$

$j=3 \quad a_5 = \frac{-2(5-3)}{4\cdot 5} a_3 = -\frac{1}{5} a_3 = \frac{4}{15} a_1$

$j=5 \quad a_7 = \frac{-2(5-5)}{6\cdot 7} a_5 = 0.$

$\Rightarrow H_5(\xi) = a_1 \xi + a_3 \xi^3 + a_5 \xi^5 = a_1 \xi - \frac{4}{3} a_1 \xi^3 + \frac{4}{15} a_1 \xi^5$

By convention, the coefficient of ξ^5 is 2^5

$\Rightarrow \frac{4}{15} a_1 = 2^5 \Rightarrow a_1 = 2^3 \cdot 15 = 8 \cdot 15 = 120$

$\Rightarrow H_5(\xi) = 120\xi^5 - 160\xi^3 + 120\xi$ (agrees with Table 2.1)

$\underline{n=6}$ (even $\Rightarrow h_{odd} = 0$)

$j=0 \quad a_2 = \frac{-2(6-0)}{1\cdot 2} a_0 = -6a_0$

$j=2 \quad a_4 = \frac{-2(6-2)}{3\cdot 4} a_2 = -\frac{2}{3} a_2 = 4a_0$

$j=4 \quad a_6 = \frac{-2(6-4)}{5\cdot 6} = -\frac{2}{15} a_4 = -\frac{8}{15} a_0$

$j=6 \quad a_8 = \frac{-2(6-6)}{7\cdot 8} a_6 = 0$

$\Rightarrow H_6(\xi) = a_0 + a_2\xi^2 + a_4\xi^4 + a_6\xi^6 = -\frac{8}{15} a_0 \xi^6 + 4a_0\xi^4 - 6a_0\xi^2 + a_0$

By convention $-\frac{8}{15} a_0 = 2^6 = 64 \Rightarrow a_0 = -15\cdot 8 = -120$

$\Rightarrow H_6(\xi) = 64\xi^6 - 480\xi^4 + 720\xi^2 - 120.$

P2.41

$$\psi(x,0) = A\left(1-2\sqrt{\frac{m\omega}{\hbar}}\,x\right)^2 e^{-\frac{m\omega}{2\hbar}x^2}$$

~~A~~ dimensionless $\xi = \sqrt{\frac{m\omega}{\hbar}}\,x$

$$\psi(\xi,0) = A\,(1-2\xi)^2\, e^{-\xi^2/2} = A\,(1-4\xi+4\xi^2)\, e^{-\xi^2/2}$$

~~But $e^{-\xi^2/2} = \left(\frac{\pi\hbar}{m\omega}\right)^{1/4} \psi_0(\xi)$~~

Recall that $\psi_0 = \alpha\, e^{-\xi^2/2}$, $\psi_1 = \sqrt{2}\,\alpha\,\xi\, e^{-\xi^2/2}$, $\psi_2 = \frac{\alpha}{\sqrt{2}}(2\xi^2-1)e^{-\xi^2/2}$

where $\alpha = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$

Thus we realize that $\Psi(x,0) = c_0\psi_0 + c_1\psi_1 + c_2\psi_2$

(because the highest power of the exponential pre-factor is 2)

Hence $\Psi(x,0) = \alpha\left[c_0 + c_1\sqrt{2}\,\xi + c_2\left(\sqrt{2}\,\xi^2 - \frac{1}{\sqrt{2}}\right)\right] e^{-\xi^2/2} =$

$$= A\,(1-4\xi+4\xi^2)\, e^{-\xi^2/2}$$

Equating <u>like</u> powers we find

Find coefficients c_0, c_1, c_2 here

$$\begin{cases} \sqrt{2}\,\alpha\, c_2 = 4A \Rightarrow c_2 = 2\sqrt{2}\,A/\alpha \\ \sqrt{2}\,\alpha\, c_1 = -4A \Rightarrow c_1 = -2\sqrt{2}\,A/\alpha \\ \alpha\left(c_0 - \frac{1}{\sqrt{2}}c_2\right) = A \Rightarrow c_0 = \frac{A}{\alpha} + \frac{1}{\sqrt{2}}c_2 = \frac{3A}{\alpha} \end{cases}$$

Normalization requires $|c_0|^2 + |c_1|^2 + |c_2|^2 = 1$

$$\Rightarrow \left(\frac{A}{\alpha}\right)^2\left(3^2 + (-2\sqrt{2})^2 + (2\sqrt{2})^2\right) = 25\frac{A^2}{\alpha^2} = 1$$

$$\Rightarrow A = \frac{\alpha}{5}$$

$$\Rightarrow c_0 = \frac{3}{5},\ c_1 = -\frac{2\sqrt{2}}{5},\ c_2 = \frac{2\sqrt{2}}{5}$$

(a) Expectation value of energy

$$E = \langle H\rangle = \sum_n |c_n|^2 E_n = \sum_n |c_n|^2\left(\frac{1}{2}+n\right)\hbar\omega =$$

$$= \hbar\omega\left(\frac{1}{2}\cdot\left(\frac{3}{5}\right)^2 + \frac{3}{2}\cdot\left(-\frac{2\sqrt{2}}{5}\right)^2 + \frac{5}{2}\left(\frac{2\sqrt{2}}{5}\right)^2\right) = \frac{73}{50}\hbar\omega$$

(b) $\Psi(x,t) = c_0\, \psi_0(x)\, e^{-iE_0 t/\hbar} + c_1\, \psi_1(x)\, e^{-iE_1 t/\hbar} +$

$+ c_2\, \psi_2(x)\, e^{-iE_2 t/\hbar}$

$E_n = \hbar\omega\left(n+\frac{1}{2}\right) \Rightarrow \frac{E_n}{\hbar} = \omega\left(n+\frac{1}{2}\right)$

$\Rightarrow \Psi(x,t) = e^{-i\omega_0 t/2}\left(\frac{3}{5}\psi_0 - \frac{2\sqrt{2}}{5}\psi_1 e^{-i\omega t} + \frac{2\sqrt{2}}{5}\psi_2 e^{-i2\omega t}\right)$

To have $\Psi(x,T) = B\,(1+2\xi)^2 e^{-\xi^2/2} = B\,(1+4\xi+4\xi^2)\, e^{-\xi^2/2}$

we need to change sign of ψ_1 term, but not of ψ_0 and ψ_2.

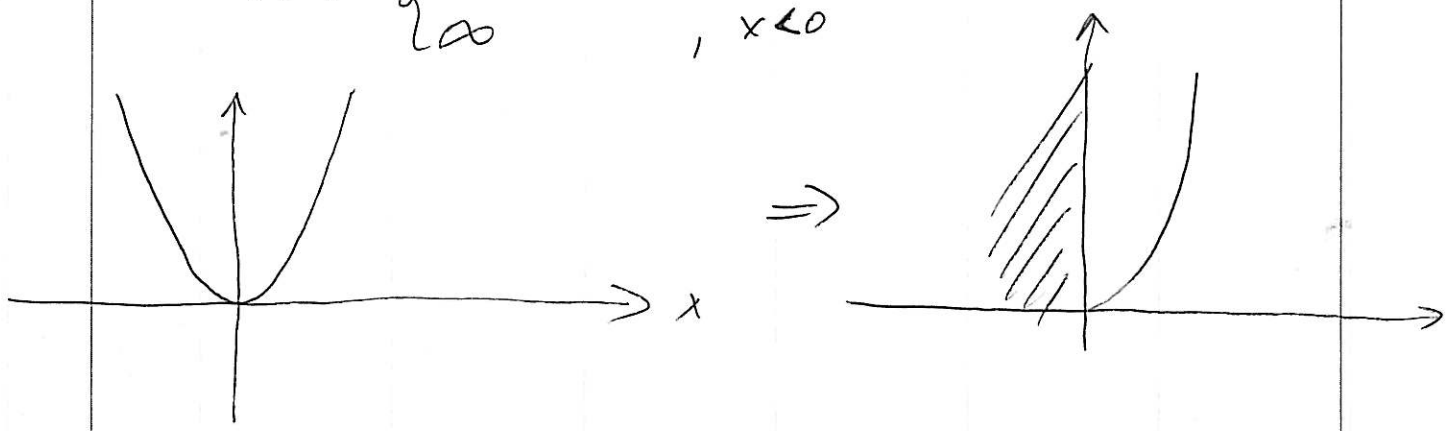
Thus $+e^{-i\omega T} = -1$ but $e^{-i2\omega T} = +1$

$\Rightarrow \omega T = \pi$ (note that $e^{-i2\omega T} = e^{-i2\pi} = +1$)

$\Rightarrow T = \frac{\pi}{\omega}$

P2.42 **Half harmonic oscillator**

$$V(x) = \begin{cases} \frac{1}{2} m \omega^2 x^2 & , x > 0 \\ \infty & , x < 0 \end{cases}$$



Clearly there is additional boundary condition now:
$\psi(0) = 0$.

Moreover, since $V = \infty$ for $x < 0$, $\psi(x<0) = 0$.
But for $x > 0$ the Schrodinger equation is the same!
$\Rightarrow$ take those solutions of the standard harmonic oscillator problem which satisfy $\psi(0) = 0$.
These are <u>odd</u> (in x) solutions: $\psi_n(x)$ with $n = 1, 3, 5 \ldots$

Corresponding energies are

$$E_{n=odd} = \hbar\omega\left(n + \frac{1}{2}\right) \underset{n=2k+1}{=} \hbar\omega\left(2k + \frac{3}{2}\right), \quad \underline{k = 0, 1, 2, 3}$$

new ground state $E_1 = \frac{3}{2}\hbar\omega$
first excited state $E_3 = \frac{7}{2}\hbar\omega$
$\vdots$